

AI Futures Trading Bot Requirements

Project Overview: I want to develop an AI-powered Futures Trading Bot that analyzes cryptocurrency futures markets and predicts the probable direction of price movement before entering a trade.

Goal: Analyze live market data; Predict UP/DOWN; Generate Buy/Sell/No Trade signals; Reduce emotional trading; Improve accuracy using AI/ML.

Timeframes: 3m, 5m, 15m, 1h.

Markets: BTC/USDT, ETH/USDT, SOL/USDT, BNB/USDT, XRP/USDT (expandable).

Each signal should include: Direction, Entry Price, Stop Loss, TP1, TP2, TP3, Confidence Score, Risk Level.

Analysis Modules: RSI, MACD, EMA, SMA, VWAP, Bollinger Bands, ATR, ADX, SuperTrend, Ichimoku, Stochastic RSI, OBV, Volume Profile, Price Action, Candlestick Patterns, Smart Money Concepts (Order Blocks, FVG, BOS, CHOCH), Market Structure, Volume Analysis, Open Interest (if available).

AI Learning: Learn from historical data, winning/losing trades, volatility, retraining.

News Analysis: Bullish/Bearish/Neutral impact from trusted crypto news.

Risk Management: Adjustable risk, max daily loss, consecutive losses, RR calculator, dynamic/trailing SL.

Dashboard: Live data, predictions, history, win rate, P/L, analytics.

Notifications: Telegram, WhatsApp, Email, Push.

Backtesting: Historical testing, win rate, profit, drawdown.

Exchange APIs: Binance Futures, Bybit, OKX, Bitget, BingX.

User Settings: Pair, timeframe, leverage, SL/TP, notifications.

Admin: Users, analytics, AI updates, API keys, health monitoring.

Suggested Stack: React/Next.js, Python FastAPI/Django, TensorFlow/PyTorch, PostgreSQL, Redis, Docker, AWS/GCP.

Security: Encrypted API keys, 2FA, audit logs, RBAC.

Performance: 1-3 second signal generation, low latency, scalable, 99.9% uptime.

Deliverables: Source code, dashboard, admin panel, documentation, deployment, maintenance option.

Important: Do not guarantee 90% accuracy or profits. Use probability-based predictions with confidence scores and continuous improvement.